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Care of the Newborn and Cord During the Third Stage of Labour (Scientific Impact Paper No. 78)

S. Bewley | R. Drain | J. Drife | A. Soe | A. D. Weeks | on behalf of the Royal College of Obstetricians and Gynaecologists

Correspondence: Royal College of Obstetricians and Gynaecologists (clinicaleffectiveness@rcog.org.uk)

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Plain Language Summary

The third stage of labour happens after the baby or babies are born and when the placenta (afterbirth) is delivered. This Scientific Impact Paper reviews new evidence related to clamping of the umbilical cord, and other interventions that impact the birthing woman and newborn during this part of labour. The paper summarises recent additional information on anatomy (structure of the circulation system) and physiology (how the system functions) of the newborn baby's circulation before the umbilical cord blood vessels close, whether naturally or after a clamp is placed on the umbilical cord.

The minutes after birth are now recognised as a time of remarkable change within the baby and we have learned that interfering with this natural process can cause potential harm. Up until birth, the fetus's oxygen supply comes from the mother via blood flow through the umbilical cord. The baby's first breath promotes rapid and profound changes to the circulation, allowing the baby to transition to a new oxygen supply from its own lungs. Fetal blood handles oxygen in a different manner to adult blood, which may be more important immediately after birth when the baby takes its first breaths. Since the last Scientific Impact Paper, more evidence confirms there is very little justification for immediate cord clamping at all gestations. Neonatal harm and death can be reduced when blood from the placenta is allowed to transfuse spontaneously to the baby in the minutes following birth. Those attending the birth should avoid intervening too early or too much while providing initial care (e.g., thermal care, stimulation). The benefits of waiting a few minutes before clamping are especially significant for premature babies.

Early cord clamping will result in the baby having a lower blood volume as it is estimated that a baby at full term will receive around 30% of its total blood volume from placental transfer via the umbilical cord. The decision on timing to clamp the umbilical cord should be based on observations of the baby, its heart rate and breathing rather than a predefined interval after birth. Also, the palpable pulsation of the cord is unrelated to the blood flowing back and forth therefore it should not be used to guide the time when clamping occurs. Clamping at the placental end allows some blood in the cord to transfer to the newborn. There are other benefits to the baby from the placental transfer of blood such as transfer of stem cells.

1 | Introduction

A previously published Scientific Impact Paper on cord clamping [1] preceded much recent evidence, which has advanced the concept of physiological cord care, underpinned by recent studies of the timing of the neonatal transition. This current paper places ‘optimal’ cord management and ‘indicated’ clamping into this context, reviews other evidence for care during transition, addresses areas of uncertainty and identifies gaps in knowledge. Knowledge of fetal anatomy and physiology is more essential than ever for clinicians observing and managing the neonatal transition.

1.1 | Language

The terms ‘early’, ‘delayed’ and ‘deferred’ cord clamping have been widely used in the literature. In this paper we try to avoid these terms as they are vague, value-laden, variously defined in research, and potentially misleading. Some health professionals use the term ‘optimal cord clamping’ but this is also baby and situation specific. In this paper, we wish to emphasise that cord clamping is a surgical procedure. As such, it must be justified on each occasion by an indication for action while the cord circulation is still functioning and the baby is transitioning from fetal to adult-type cardiorespiratory function.

2 | New Information on the Neonatal Transition

The transition from fetus to neonate is the most rapid and fundamental physiological change a human body undergoes in its lifetime. It is technically difficult to study but recent research has improved our understanding of this remarkable phenomenon.

2.1 | Anatomy

When the baby's first breaths expand the lungs, the dramatic decreases in intrathoracic pressure and pulmonary vascular resistance enable blood flow from the right ventricle to the lungs, lowering the right ventricular pressure, and increasing blood volume and pressure in the left atrium. Blood returns to the left atrium, where the reverse in the pressure gradient between the left atrium and right atrium immediately closes the foramen ovale [2]. This functional closure is followed later by tissue fusion [3].

The ductus arteriosus (DA) is kept open during pregnancy by prostaglandins produced mainly by the placenta [4]. After birth, loss of the placental circulation results in a reversal of flow through the DA, which then begins to contract, with closure usually complete within 7 days [5].

A third shunt, the ductus venosus (DV), takes blood from the umbilical vein to the inferior vena cava during fetal life. Inspiration preferentially directs blood flow from the DV into the right atrium, possibly driving placental transfusion [6], and it closes during the first week of life in term infants.

2.2 | Physiology

2.2.1 | Breathing

Before birth, the baby makes breathing movements intermittently [7]. These movements are suppressed by hypoxia [8] and raised prostaglandin levels [9]. At birth, diffuse tactile and cold stimuli, as well as changes in blood gas concentrations, result in the rapid onset of vigorous breathing. When the neonate's first breaths expand the lungs, there is a rapid reduction in pulmonary vascular resistance. Aeration of the alveoli drives fluid from the air spaces into the perialveolar tissue, where it is cleared by the lymphatics and blood vessels [10].

2.2.2 | Pulmonary Blood Flow

At birth, pulmonary blood flow quickly increases. In most infants the flow in the DA ceases within 48 h [5]. Left ventricular output doubles within an hour of birth, with a concomitant rise in stroke volume. There is a rapid and transient rise in cerebral blood flow [11]. Jain et al. [12], using echocardiography, found that all indices of right ventricular function increased, and cardiac output almost doubled soon after birth. Just before birth, cardiac output from the two ventricles combined is about 450 mL/kg/min (about 1.5 L/min for a 3.5 kg baby) with two-thirds coming from the right ventricle and almost 80% of the blood bypassing the lungs. After birth, when 100% of the cardiac output flows through the lungs, each ventricle has to pump about 400 mL/kg/min.

These studies quantified blood flow but not the total amount of blood involved. The average weights of the placenta and the lungs at term are 508 g and 40 g, respectively, but the volumes of their vasculature are hard to measure. Acharya et al. [13] reviewed the literature and calculated that the total volume of blood in the fetus and placenta is 110–115 mL/kg body weight—i.e. almost 400 mL for a fetus weighing 3.5 kg at term. The average placenta at term contains 33% of the total (about 130 mL of blood); after normal blood transfer this reduces to 13% of the total (around 50 mL) [14].

The blood itself changes after birth. Fetal haemoglobin (HbF) has a high oxygen affinity and releases oxygen more readily than adult haemoglobin (HbA) in response to a small decrease in arterial partial pressure of oxygen. This may be particularly important for its benefit during the minutes after birth before the neonate takes its first breath. After birth, production of HbF diminishes and normal levels of HbA are achieved at 4–6 months of age [11].

2.2.3 | Placental Separation

In an observational study of 151 women following vaginal birth, where 99% had a physiological third stage of labour and only four had a postpartum haemorrhage (PPH) and received uterotonic drugs for PPH treatment, Padwardhan et al. [15] reported a mean third stage of labour duration of 8 min (range 2–39 min) and that 80% of the women delivered their placenta within 12 min.

2.2.4 | Umbilical Cord Blood Flow (Arterial and Venous)

During and after the third stage of labour, the baby continues to pump blood into the placenta. Boere et al. [16] put a Doppler ultrasound probe midway along the cord to study umbilical blood flow after vaginal birth before cord clamping (between 4 and 9 min). Arterial flow was sometimes bidirectional and venous flow stopped or even reversed when the baby cried hard. In 17% of the 30 infants there was no arterial flow, in 40% it stopped at a median of 4 min 22 s and in 43% it continued until cord clamping. Venous flow showed a similar pattern. In 10% there was none, in 57% it stopped before the cord was clamped (at a median duration of 4 min 34 s), and in 33% it continued until clamping.

Importantly, the cord blood flow was unrelated to the cessation of pulsation. The pulse is a pressure wave in the vessel wall, not in its content. Boere et al. [16] recommended that pulsation should not be used to guide the time of cord clamping. This finding casts doubt on studies of neonatal outcome based on observations of cord pulsation (e.g., Wu et al. [17]).

The most recent estimate of neonatal blood volume in 2024 (85–95 mL/kg) [18] supports what we already know about blood transfer through the cord (placental transfusion).

2.3 | Impact of Interventions During the Third Stage on Transition

Physiological transition takes some time and early clamping interrupts this vital process.

2.3.1 | Uterotonic and Other Maternal Risks

Transfer of oxytocin across the placenta is very slow [19] and clinical studies show that newborn cord oxytocin comes from the fetus, not the mother [20]. Thus, uterotonic administration does not need to be delayed and can be administered with the cord intact. Classic studies showed that uterotonic administration at birth appears to accelerate the rate of placental transfusion with an intact cord, but with no effect on the final volume [21]. Randomised controlled trials (RCT) have found no evidence that the timing of cord clamping affects PPH risk at vaginal [22] or caesarean births [23, 24].

2.3.2 | Clamping

After interrupting the circulation, blood is retained in the placenta. An experiment using full term, in utero placentas with immediate clamping after 10 IU prophylactic oxytocin showed that up to 80% of the residual volume can be collected (with or without cord milking) in the first 60 s following birth if the clamp is released. The amount collected was less when a low dose intravenous oxytocin infusion was used at caesarean birth [25]. Although results from disconnected placentas cannot be directly applied to clinical practice, this aligns with the experience with deferred cord clamping.

2.3.3 | Impact on Venous Return to the Neonate's Left Atrium

Clamping the cord before sufficient lung aeration is achieved reduces the pre-load to the left side of the heart, which affects and reduces cardiac output [26, 27], immediately leading to bradycardia in the newborn [28]. In turn, this can lead to inadequate tissue perfusion as a newborn's shock response preferentially supplies vital organs at the expense of non-vital tissue.

2.3.4 | Gravity

Gravity affects the rate of placental transfer when the baby is held at 40 cm or more above or below the placenta [29]. The effect is not seen at just 20 cm, nor when a baby is placed on a recumbent mother's abdomen after birth [30]. In some clinical trials of premature babies, holding the baby below the level of the mother's uterus has been used as a strategy to speed up the rate of transfer [31].

2.3.5 | Cord Milking

Milking of the cord prior to clamping achieves similar haematological changes for the neonate as deferred cord clamping, but in a very short time scale [32]. Even though milking is associated with better outcomes than immediate cord clamping, this is no longer the right control comparison [33], and a justification for using it must be made and documented (i.e., why immediate clamping was necessary). There are no data about the impact on neonatal hypoglycaemia [34]. However, milking does not appear to result in reduced mortality in premature babies in the way that deferring clamping for at least 2 min does [35]. Indeed, recruitment of babies under 28 weeks of gestation in the PREMOD2 study was stopped early due to an excess of severe intraventricular haemorrhage (IVH) compared with clamping at 60 s [36]. It seems inappropriate to recommend the routine use of milking given the demonstrated harm in the very premature (which may be present but more subtle and at lower rates in older babies), the speed of transfusion with milking and the availability of a physiological alternative (variously called 'watchful waiting', 'leaving well alone' or 'doing nothing skilfully').

In term neonates with immediate cord clamping, around 25 mL of blood (or 8 mL/kg) is retained within each 30 cm of cord [37]. Thus, whatever remains of this blood at the time of clamping can be made available to the baby if clamps are placed at the placental rather than neonatal end of the cord.

2.3.6 | Stem Cells

Placental blood is an abundant source of stem cells, which have an important potential role in repairing any organ injury the baby might sustain [38, 39]. The volume of retained placental blood depends heavily on third stage practices. Stem cell donation has long been of interest to third parties [40–42] and stem cell therapy from umbilical cord blood collection is being explored as treatment for hypoxic-ischaemic encephalopathy [43]. This is another important aspect to providing maximal placental blood to the newborn through 'optimal' umbilical cord management [44].

3 | Re-Examining the Rationale for Cord Clamping and Its Timing

Accepted practice for the neonatal intervention of umbilical cord clamping during transition has changed over time. Routine clamping became standard practice within active management of labour protocols aimed at preventing PPH [45] but it is long recognised that it should be deferred [1].

In April 2025 PubMed searches for 'delayed cord clamping' and 'placental transfusion' yielded 1531 and 3265 papers respectively, and in both groups two-thirds of the publications were from the decade 2013 to 2023. Some were uncontrolled studies of neonatal outcomes and some listed benefits of 'delayed' cord clamping based on scanty evidence. Others were systematic reviews and guidelines on the timing of cord clamping in term and preterm births. Recent systematic reviews show that 'delayed' clamping is preferable to 'immediate' clamping in term and preterm neonates, concurring with earlier Cochrane reviews [33, 46–48].

3.1 | Preterm Birth

The benefits of waiting before clamping are clearest in preterm babies of less than 34 weeks of gestation, as the background rates of serious events are higher. A study showing a 32% decrease in mortality with 'deferred' rather than 'immediate' clamping [48] has been confirmed with high-level certainty in the latest meta-analysis of trials [35]. These analyses were highly influenced by the largest of the studies, an international multicentre RCT of 1566 births where the cord was routinely kept intact for the first 1 min after birth, with newborns receiving nothing more than drying and stimulation during this time [31].

The meta-analyses included some studies where preterm infants received intact cord resuscitation [35]. The introduction of bedside resuscitation trolleys allowing cord-intact resuscitation meant that much longer times to clamping became possible, but these were rarely analysed separately in study data. An individual participant analysis in 2023, however, allowed examination of the data according to duration. This revealed a 69% reduction in mortality when the cord remains intact for over 2 min [33]. The reason for this dramatic effect is unclear but may be because abrupt clamping of the umbilical arteries before lung aeration causes harm through a hypertensive surge and reduced cardiac output in the baby [49].

While further research is clearly needed, there is a physiological and clinical rationale to offer preterm and term infants that appear to require resuscitation, initial airway manoeuvres and inflation breaths with an intact cord for at least 2 min for preterm infants. This can be followed by cord clamping and further resuscitation on a resuscitation platform if still required. Immediate cord clamping and removal of the baby to the resuscitation equipment may be unnecessary and make some babies worse.

3.2 | Term Birth

In term infants, harms and benefits are harder to measure because of lower background rates of severe complications. In a systematic review of mixed gestation, 'delayed' cord clamping was associated with a 27% reduction in neonatal mortality, an effect consistent across all gestational ages (15 studies, 3041 infants, RR = 0.73, CI: 0.55–0.98, $p = 0.03$, $I^2 = 0\%$) [34]. Delayed clamping increases neonatal blood levels of ferritin, a critical ingredient for neurone formation [50], which may improve brain development. This was suggested by a systematic review of four studies from three RCTs examining neurodevelopment at 4 months of age—most, but not all, being improved with delayed cord clamping [51].

The largest randomised trial with 68% four-year follow-up of babies receiving either less than 15 s or greater than 180 s clamping with gases taken with an intact cord, [50, 52] noted: "Delayed cord clamping results in better motor and social development at 4 years of age especially in boys", and recommended the wait should be: "At least 3–5 minutes", although the upper limit is more theoretical.

3.3 | Other Plausible Neonatal Benefits of Avoiding Intervention During Transition

From studies with serial intact cord blood gas collection and RCTs, placental acid–base and gas exchange may continue after birth with an intact cord [53], possibly allowing a degree of acidosis correction. Near-infrared spectroscopy is also being used to study cerebral oxygenation [54] but the evidence is conflicting.

While it is worrying that immediate cord clamping would halt any natural correction of the hypoxia/acidosis of labour (recovery or so-called autoresuscitation) and may be harming brain oxygenation, it is not yet possible to extrapolate or quantify the impact on sick babies. Just 30 s more placental transfusion at caesarean birth allowed a rise in CD34+ cell count (a marker of haemopoietic and multipotent lineage stem cells with their regenerative and repair potential) [55, 56].

3.4 | Physiological Based Cord Clamping (PBCC)

The uncertainty over the exact timing of cord clamping is built into an approach [10] that was first studied in preterm lambs [49] and then humans [57]. Despite the misnomer of 'physiological' intervention, the principle is clear: umbilical venous return should be allowed to continue until pulmonary blood flow is fully established. If the cord is clamped before ventilation, inadequate venous return from the lungs could put the infant at risk of asphyxia and also an ischaemic insult due to low cardiac output. Timing of cord clamping should, therefore, be based on the infant's physiology or condition, not an arbitrary set period. Hooper et al. [58] hypothesised that early clamping may be of greatest harm to apnoeic infants and this has since been demonstrated in randomised trials [33]. Therefore, PBCC is defined as clamping only after the infant has started breathing (or has received respiratory support) and the lungs have been aerated. No specific time is mandated.

4 | Neonatal Transition: How and When to Manage the Cord

Since the earliest development of the specialty of neonatology, babies have been removed from the mother to be taken to a special platform that enables skilled specialist resuscitation with oxygen, monitoring, a heater, clock and other equipment. However, the need for this urgent separation is challenged by the evidence.

4.1 | Birth at Term and Preterm

Clinicians should already be aware of the widespread agreement about the benefits of avoiding too-early clamping of the still functional umbilical cord. This followed recommendations from the National Institute for Health and Care Excellence (NICE) [59] (at least 'a minute'), the previous RCOG Scientific Impact Paper [1] ('deferred') and the World Health Organization (WHO) advice [60] ('late' 1–3 min cord clamping in well newborns). Avoiding harmful intervention may also avoid unnecessary resuscitation efforts, although it is unknown if there is too long a wait.

4.1.1 | New Guidance

There have been several updates to clinical advice in the last decade [44, 59, 61–65]. The British Association of Perinatal Medicine's updated guidance and Quality Improvement Toolkit on Optimal Cord Management in Preterm Births [44] before 34 weeks of gestation is a detailed practical resource on how to avoid barriers and achieve effective implementation and safety. Two studies of optimal cord management observed that 90% or more of all included babies started to breathe before 60 s, so the majority will not need additional respiratory support during the first minute of optimal cord management [66, 67]. The Resuscitation Council UK published Newborn Resuscitation and Support of Transition of Infants at Birth Guidelines in 2025 [63].

4.1.2 | Current Practice

It is not known whether or how practice has evolved despite documenting the time of cord clamping being a NICE Quality Standard since 2014 [68]. A local quality improvement initiative in Australia from 2014 to 2022 [69] showed immediate clamping in preterm neonates before 34 weeks of gestation falling from 88% to 30%, hypothermia falling from 76% to 43% and documentation improving from 16% to 93%. A US single-centre, before-and-after study of changing from waiting at least 60 s to between 120 and 180 s (in those infants breathing by 60 s of life) for neonates of less than 33 weeks of gestation showed reduced mortality [70]. Data from the National Neonatal Audit Programme (NNAP), which adopted optimal cord clamping within a bundle of measures in 2019, shows practices are changing for premature babies in the UK with over 60% of babies born at under 34 weeks of gestation now receiving deferred cord clamping of at least 1 min [71]. In Sweden, which has recommended more than 2 min since 2008, professionals overwhelmingly recognise the timing of clamping as important in neonatal care [72], and its observed median in vigorous infants is 6 min [73, 74].

Studies examining initial stabilisation during optimal cord management include widely varying methodologies; firm conclusions cannot yet be made regarding safety or benefit. However, some units now routinely undertake initial stabilisation with the cord intact and have reported favourably on feasibility and safety of platforms (e.g., a bedside trolley or platform) to assist this [47, 75–77].

4.2 | The Newborn Requiring Resuscitation

Immediate cord clamping is not essential for resuscitation, nor for proper assessment of the non-vigorous baby and may even compromise resuscitation. Improved outcomes have been reported with intact cord resuscitation, although the waiting time comparators vary. There are ongoing trials of stabilisation and resuscitation of preterm, term and sick babies (e.g., Hutten et al. [78]).

A study from Nepal of 230 newborn infants [79], found improved immediate outcomes after intact cord resuscitation compared with cord clamping at less than 1 min. Raina et al. [80], in a study from India of 496 babies at 34 weeks of gestation or above, compared cord clamping at less than 30 s with intact cord resuscitation on a bedside resuscitation trolley and found that the intact cord group had less need for ventilation and improved Apgar scores and oxygen saturation. These results contrast with those from high income settings where little benefit is seen for cord-intact resuscitation. At later gestational ages, PBCC has been compared with early clamping and transfer to a side-room resuscitation unit. The BabyDUCC study in 123 babies of greater than 32 weeks of gestation with delayed breathing found that cord-intact ventilation until breathing established had no clinical benefit over early cord clamping and side-room ventilation (median clamping time 136 versus 37 s) [81].

Similarly, in premature babies born at under 30 weeks of gestation in the ABC study, there was no difference in intact survival rates in 669 infants randomised to PBCC or clamping at 30–60 s with standard side-room resuscitation. However, the study found reduced rates of sepsis and blood transfusions, and improved survival for male babies. Parents were less anxious and more content in the PBCC arm [82]. In the VentFirst study, 548 babies at under 29 weeks of gestation received either cord-intact resuscitation with cord clamping at 120 s, or cord clamping at 30–60 s and side-room resuscitation. No effect was seen on death or IVH [83]. Both the ABC and VentFirst studies had a relatively large number of babies that were breathing and did not need rescuscitation. Babies who are well may respond differently to those who are hypoxic and the latter may benefit from the oxygen, blood and stem cells that could be obtained from the umbilical cord. As sick and well babies were included, the outcome comparisons were underpowered for sick babies and further research is needed.

A randomised trial (PREMOD2) of delaying for 1 min versus cord milking found an increase in severe IVH in babies under 28 weeks of gestation with milking; these babies were excluded from the later parts of the study [36]. At later gestations, there were no differences in rates of severe IVH and death. The same effect was not seen in the PCI study of 209 babies under 30 weeks of gestation randomised to cord milking versus intact cord resuscitation for 3 min [84].

4.3 | Can a Baby Receive Its Placental Transfusion and Be Cared for Away From the Mother?

Ideally, the baby would receive its placental transfusion and stay close to the mother. If it is necessary to remove a baby at some point after assessment, partial transfusion can be achieved by clamping and cutting the cord at the placental rather than neonatal end, enabling residual cord blood to enter the baby. Achieving greater blood transfusion and removing the baby requires early delivery of the placenta and then transfer of both the placenta with the baby to the neonatal team. At caesarean birth, a placenta can be manually delivered while still attached to the baby, allowing the operation to continue while avoiding disrupting the placental transfusion [85]. Alternatively, extremely premature babies can be delivered inside the intact amniotic sac (en caul) with the placenta. One RCT involving 60 cases of premature babies (born later than 23 weeks of gestation but weighing less than 1500 g) delivered by caesarean birth begins to address this. It compared clamping at 30–60 s versus delivering the baby, intact cord and placenta together, for extrauterine placental perfusion [86]. Babies weighed 970 and 982 g on average (1% heavier but a more sizeable increase in blood volume). Oxygen perfusion was better in the latter group although there was no difference in haematocrit or neonatal parameters. Nevertheless, the study demonstrates feasibility.

4.4 | Special Situations

4.4.1 | Rhesus Alloimmunisation

Traditionally, early cord clamping was preferred because rhesus (Rh) alloimmunisation remains a cause of neonatal hyperbilirubinaemia, neuromorbidity, and late-onset anaemia. However, an RCT of Rh-alloimmunised infants of 28 to 41 weeks of gestation comparing early (less than 10 s) cord clamping ($n = 34$) versus more than 60 s clamping ($n = 36$) showed the latter babies had a higher mean packed cell volume at 2 h of age (primary outcome) [87]. Avoiding early clamping improved the haematocrit, resulted in better haemodynamic stability and decreased the need for transfusion in early infancy with no increase in the incidence of serious side effects, such as double volume exchange transfusion and hyperbilirubinaemia. This has refuted theoretical concerns that allowing placental transfusion is harmful.

4.4.2 | Multiple Pregnancy

While clinicians report being comfortable deferring cord clamping in dichorionic twins, surveys suggest that practice varies in monochorionic twins, with concerns about the possibility of acute twin-to-twin transfusion [88]. Meta-analysis of randomised studies including over 2000 premature infants are reassuring for dichorionic pregnancies, showing deferred clamping reducing mortality and transfusion, as in singletons [89]. There is also reassuring data from studies of monochorionic twin pregnancies. Yoon et al. [90] retrospectively reviewed 107 monochorionic twin births (214 neonates) and found improved haemoglobin levels in the 62 newborns who received deferred cord clamping compared with those with immediate cord clamping. In the analysis of 1410 newborns of all chorionicity, there were significantly fewer respiratory complications, neonatal unit admissions and blood transfusions in those with delayed cord clamping but this was not analysed by chorionicity. A randomised trial of preterm multiple births did not provide separate data for monochorionic pregnancies, but there were no adverse effects seen in the 55 births [91].

There are no data to direct recommendations in monochorionic twin pregnancies with complications related to vascular connections in monochorionicity (i.e., twin anaemia-polycythaemia sequence [TAPS] and twin-to-twin transfusion syndrome [TTTS]) where there is the potential for acute transfusion events at birth. There is one report of three cases of preterm monochorionic twins with antenatal suspected severe anaemia of the donor fetus, which demonstrates feasibility of deferred cord clamping for the donor infant with immediate clamping of the recipient cord [92]. Further studies are required to demonstrate benefit.

4.4.3 | Severe Growth Restriction

Several small studies have been conducted in growth restricted babies and have found improved haemoglobin in the neonate without adverse effects [93, 94].

4.4.4 | Nuchal Cord

To avoid pre-birth or immediate cord clamping when there is a tight cord around the neck, a 'somersault' manoeuvre usually enables delivery with an intact cord [95]. Many images and videos of this simple technique are available online (see resources in Appendix A).

4.4.5 | Out of Hospital and Unplanned Births

Planned births at home or in midwifery units usually enable keeping the baby close and avoiding clamping. Newborn resuscitative measures can be achieved with an intact cord, although the baby may be removed to a firm, flat surface. Low-cost platforms are in development to facilitate bedside resuscitation [96].

In unplanned births, often attended by ambulance services and dependent on the environment, the cord can be kept intact until it is empty/white, so long as safe handling and thermoregulation can be achieved after drying and keeping the baby warm.

4.4.6 | Antepartum and Postpartum Haemorrhage

Maternal vaginal bleeding or abruption may be difficult to manage but neither is an indication for early cord clamping. Haemorrhage of fetal cells into the maternal circulation or vagina is rare even with confirmed abruptions and would also be less than the usual 90 mL of placenta-neonatal autotransfusion. In one series of 68 confirmed abruptions, only four had any evidence of fetal cells in the maternal circulation [97]. Where fetomaternal haemorrhages do occur, the vast majority are under 4 mL, with only 0.03% being over 15 mL [98]. There is no evidence in the literature about fetal bleeding from the placenta or cord vessels (e.g., incising the placenta at caesarean birth, bleeding vasa praevia) although a bleeding, detached cord should be immediately clamped. In the event of rapid maternal vaginal haemorrhage during the third stage where the placenta is not delivering with cord traction or simple manual extraction, it would also be prudent to clamp the cord before piecemeal placenta removal.

4.4.7 | General Anaesthetic Caesarean Birth for Acute Fetal Concern

Both induction and inhaled anaesthetic agents rapidly cross the placenta and there has been concern about resulting respiratory depression in the newborn. However, a systematic review of 29 randomised trials showed no difference in Apgar scores, cord pH or need for oxygen resuscitation between those who have general and regional anaesthesia [99]. Given the study dates, most babies in the review were likely to have had immediate cord clamping, which could have reduced the neonatal effects. However, there are reassuring data from a matched cohort of 60 babies in which there was a greater than 10-min delay from induction of anaesthesia to birth. Apgar scores at 1 and 5 min were lower in the neonates born after general anaesthetic; however, cord blood pH values, neonatal intensive care unit admission and length of stay were similar in the two groups [100]. Given this, it would seem unnecessary to rush into immediate cord clamping after general anaesthetic caesarean birth while the baby is still receiving the oxygenated placental blood.

4.4.8 | Neonatal Bedside Resuscitation With an Intact Cord

Babies need to be properly assessed, dried and stimulated, and simple actions such as airway manoeuvres and inflation breaths can be performed while the cord is intact. However, if bag and mask oxygenation is still required after 1 min of stimulation and bedside resuscitation is unavailable, then it would be appropriate to clamp and cut the cord to move the baby to a resuscitation area.

4.4.9 | Fetal Anomalies

Current advice regarding antenatally diagnosed diaphragmatic hernia is to avoid early clamping [101]. This is supported by experimental animal work showing cardiopulmonary benefits [102] but results from an ongoing multicentre trial are pending [103].

There is more caution and less consensus for cardiac anomalies. Given its major impact on the newborn's cardiac pressures during transition (see Section 2), early cord clamping is a potentially dangerous intervention in infants with cardiac abnormalities. Robust outcome data are lacking but clinicians have expressed willingness to conduct randomised trials [104].

Babies with congenital anomalies undergoing corrective surgery often require blood. Studies have shown the possibility of still collecting over 50 mL of blood after birth to save for autologous transfusion with ‘delayed’ (45 s) clamping, but this practice is unsupported by clinical outcome evidence [105]. The British Society of Haematology now advocates against pre-deposit autologous blood donation [106] and most hospitals would not be able to store the blood as per regulatory advice [107].

4.5 | Does Bedside Resuscitation Cause Distress?

4.5.1 | The Mother and Carers

Parents' views of bedside neonatal care are largely positive [82, 108]. However, there can be some negative emotions and anxiety about witnessing resuscitation; clear information is crucial [109].

4.5.2 | Clinicians

Clinicians report that giving bedside neonatal care improves communication with parents, but some (especially inexperienced clinicians) express concerns about performing intensive interventions in front of them [110, 111]. There are practical difficulties to overcome [112] – multiprofessional training is important [110].

4.6 | Challenges to Implementing Change

Barriers to changing practice have been described [113] and a formal quality improvement project is usually required to change ingrained practices, especially around neonatal resuscitation. A high profile, multidisciplinary campaign with senior support and local champions is required to initially change practice and then to maintain it [114]. A checklist mnemonic has been suggested as an aide-memoire to overcome resistance to changing entrenched third stage care practices. BOND stands for Baby skin-to-skin, Oxytocic, iNitiate blood loss monitoring, and Delay (or Don't Do) cord clamping [115].

The difficulty in achieving change is demonstrated by a 2025 survey of maternity units in England and Wales, which found that while all but one of 116 were practising deferred cord clamping for uncomplicated term births, 30% of units did not defer cord clamping for premature births at all. Furthermore, cord intact stabilisation was used in planned and emergency births by only 21% and 3% of units, respectively [116]. The underlying reasons given concerned difficulties with bedside resuscitation equipment. However, cultural inertia, fear of causing harm in a high-risk situation and uncertainties about the respective roles of obstetricians and neonatologists may have also contributed.

There have been concerns about how to obtain cord blood for pH assessment following deferred cord clamping. Wiberg et al. [117] showed how paired samples could easily be collected from an intact cord, and that delaying the collection results in a slow acidotic change. Double clamping of a cord segment stabilises the values for 20 min [118] but there is little blood left for analysis once the cord empties during deferred cord clamping. The simplest solution is to collect paired samples from the intact cord, recording carefully the exact time of both cord clamping and sample collection in case this is required legally.

Extensive data sets have been developed using blood collected from intact cords with the placenta still in situ, without risk to the neonate [119]. The prospect of serial blood gases being taken with the cord intact might aid individual clinical care and guide further research into resuscitation.

5 | New Research Avenues for Continuing Uncertainties

Understanding of fetal-to-neonatal transitional anatomy and physiology has increased but further research is needed on the long-term impacts of many interventions undertaken while dramatic changes are, or are not, taking place in the first minutes of life during this transition or as part of neonatal resuscitation.

The National Institutes of Health Research (NIHR) is applying artificial intelligence models to large data sets to help predict and prevent brain injury. Extreme caution should be applied, as relevant details of cord clamping and its sequencing with breathing will largely be missing from records [120].

Animal work investigating resuscitation after various times of clamping is required, partly as continuous fetal heart monitoring in labour has not been shown to prevent death and brain damage compared with intermittent auscultation [121, 122]. Connecting delivery, transitional and postnatal observations, with and without intact cord resuscitation of sick and asphyxiated newborns, should generate hypotheses and novel approaches to recovery.

Birth in the community, nuchal cord, fetal conditions of concern and vaginal bleeding must be further explored in future research as avoiding unindicated clamping finds its place in improving rare but devastating outcomes.

New ideas such as minimally invasive monitoring through both second and third stages of labour hold promise (e.g., a prototype sensor for continuous fetal lactate monitoring that measures a biological variable closer to the brain-damaging target [123]).

Implementation research worldwide focuses on multiprofessional training strategies as knowledge alone is not enough to change practice [124–127].

Uncertainties will be lessened with more evidence coming from ongoing trials on the benefits and harms of cord-intact resuscitation.

6 | Opinion

Current evidence suggests:

6.1 | General Principles

- Third stage focus should move from clock-watching to mother-and-baby-watching.
- All third stage interventions should be evidence-based and indicated.

6.2 | Treatments, Procedures and Interventions That Are Not Appropriate

- Routine interruption of fetal-to-neonatal transition has no role.
- Immediate umbilical cord clamping before lung aeration is detrimental and contraindicated. If used, the clinician must document why waiting was not feasible.
- Cord clamping should be avoided for at least 2 min in preterm births.
- Cord clamping should be avoided for at least 1 min in term births, with good theoretical reasons for offering at least 2 min.
- Cord milking should not be performed routinely.

6.3 | Practical Implications

- Cord function and closure should be explained to parents as part of information about the baby's transitional state during the third stage of labour.
- The indication and timing of cord clamping should be documented in medical notes in completed minutes and seconds and in relation to heart rate and establishment of breathing.
- Cord-intact bedside resuscitation has both a physiological and clinical rationale for those preterm and term infants that require airway manoeuvres and inflation breaths. It is feasible, safe and reduces anxiety in parents. However, it is not yet clear whether this further improves clinical outcomes beyond that gained by drying and stimulating with the cord intact for 1 min and then cutting for transfer to side-room resuscitation.
- Clinical staff need to be aware of current guidance around cord management.
- More education and training are required for maternity staff who are accustomed to handing over the baby immediately to neonatal practitioners.
- Units that have solved and overcome obstacles to change should share their good practice learning with units initiating or undergoing the process.

6.4 | Implications for Research and Policymakers

- Strategies and equipment for providing initial cord-intact care at the woman's bedside should be developed further and evaluated.
- Healthcare systems of oversight (training, audit, governance) need an understanding of how assessments and current standards are being performed.
- Electronic patient records may need to be adapted to note exact timings and sequencing, especially for babies that experience resuscitation.
- Further research is needed on how long the cord should remain intact for fetal-to-neonatal transition care in premature babies, during resuscitation for babies born in poor condition, and with specific conditions.

Inclusive Language

This guidance is for healthcare professionals who care for women, non-binary and trans people. Within this document we use the terms woman and women's health. However, it is important to acknowledge that it is not only women for whom it is necessary to access women's health and reproductive services in order to maintain their gynaecological health and reproductive wellbeing. Gynaecological and obstetric services and delivery of care must therefore be appropriate, inclusive and sensitive to the needs of those individuals whose gender identity does not align with the sex they were assigned at birth.

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Declarations of Interest

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Appendix A

Extra Resources for the Public

Support and material may be found from groups such as:

- Tommy's: https://www.tommys.org/pregnancy-information/giving-birth/delayed-cord-clamping-optimal.
- Twin Anaemia Polycythemia Sequence Support Foundation: https://www.tapssupport.com.
- Wait for White: https://waitforwhite.com/.
- Blood to Baby: https://www.bloodtobaby.com/.

Extra Resources for Clinicians

- Diaz-Rossello JL, Blasina MF. Care of the Newborn Infant during the Third Stage of Labor. In: Moreira de Sá RA, Borges da Fonseca E. (eds) Perinatology. Springer, Cham. 2022. https://doi.org/10.1007/978-3-030-83434-0_53.
- Maternity and Newborn Safety Investigations (MNSI). 2025. Webinar series 2: Umbilical Cord Management WEBINAR SERIES 2: Exploring learnings from MNSI safety investigations or https://youtu.be/wCm-d8vbpgA.
- Somersault manoeuvre shown in videos such as: https://www.youtube.com/watch?v=VIyHZyij0Bg and https://www.msdmanuals.com/professional/multimedia/video/cord-around-the-neck.

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